Conditional Access Integration

Implementation Solution Guide

Version 1.0.0 · 2026-06-13

Contents

Executive Summary — Conditional Access Integration (Microsoft Intune) 4

What you get . . . 4

What this delivers . . . 4

The standard deployment model . . . 4

Outcome . . . 4

1. Solution Overview — Conditional Access Integration 4

What Conditional Access is . . . 4

How it integrates with Intune . . . 5

How it works . . . 5

The baseline policies this delivers . . . 5

Benefits . . . 5

Limitations (be honest with buyers) . . . 5

The standard deployment model . . . 5

2. Requirements Guide — Conditional Access Integration 6

2.1 Licensing . . . 6

2.2 Break-glass (emergency access) accounts . . . 6

2.3 Device compliance (for "require compliant device") . . . 6

2.4 MFA registration . . . 6

2.5 Sign-in logs / reporting . . . 6

2.6 App protection (optional — mobile) . . . 6

Checklist . . . 7

Global Requirements — Intune Solutions (category-level) 7

G1. Microsoft Entra ID tenant . . . 7

G2. Microsoft Intune subscription . . . 7

G3. Administrative roles . . . 7

G4. Device enrollment in Intune . . . 7

G5. Pilot security group . . . 7

G6. Supported Windows baseline . . . 8

G7. Network access to Microsoft cloud endpoints . . . 8

3. Implementation Guide — Conditional Access Integration 8

Step 1 — Create and exclude break-glass accounts . . . 8

Step 2 — Create the pilot group . . . 8

Step 3 — Policy: require compliant device for Microsoft 365 . . . 8

Step 4 — Policy: require MFA . . . 9

Step 5 — Policy: block legacy authentication . . . 9

Step 6 — (Optional) require app protection for mobile . . . 9
Step 7 — Review Report-only impact . . . 9
Step 8 — Enable enforcement . . . 9
Step 9 — Expand to production . . . 10
5. Testing Guide — Conditional Access Integration 10
Test 1 — Policies evaluate in Report-only . . . 10
Test 2 — Compliant device satisfies the control . . . 10
Test 3 — Noncompliant device is challenged/blocked . . . 10
Test 4 — MFA is required . . . 10
Test 5 — Legacy authentication is blocked . . . 10
Test 6 — Break-glass accounts are exempt . . . 10
Test 7 — Policy + sign-in report . . . 10
Pass/fail gate . . . 11
6. Troubleshooting Guide — Conditional Access Integration 11
A. Lockout risk / admins blocked . . . 11
B. "Require compliant device" blocks legitimate users . . . 11
C. MFA issues . . . 11
D. Legacy auth still working . . . 12
E. Report-only shows nothing . . . 12
F. Diagnostics quick reference . . . 12
G. When to escalate . . . 12
7. Best Practices — Conditional Access Integration 12
8. Deliverables Summary — Conditional Access Integration 13
Documentation . . . 13
PowerShell scripts . . . 13
What the buyer receives . . . 13
Outcome . . . 13
Appendix A — PowerShell Scripts 14

Executive Summary — Conditional Access Integration (Microsoft Intune)

White-label · Intune Solutions

Device compliance and MFA only protect you if access is actually *gated* on them. **Conditional Access** is the enforcement engine: it evaluates every sign-in against your rules — is the device compliant? has the user passed MFA? is this legacy authentication? — and allows, challenges, or blocks accordingly. Integrated with Microsoft Intune, it turns device compliance into a hard access requirement.

This is a complete, **deploy-it-today** package for implementing Conditional Access integrated with Microsoft Intune, written for an administrator with no prior Conditional Access experience — including the safety practices (break-glass accounts, Report-only mode) that prevent lockouts.

What you get

- A plain-language **Solution Overview** of how Conditional Access evaluates and enforces sign-ins.
- A **Requirements Guide** with verify / implement-if-missing for every prerequisite.
- An **Implementation Guide** of precise, click-by-click steps in the required order — Report-only first.
- Five **PowerShell scripts** for device-state discovery, access validation, policy reporting, sign-in analysis, and break-glass verification.
- **Testing** and **Troubleshooting** guides for confident, lockout-free rollout.

What this delivers

A baseline Conditional Access posture: **require a compliant (or Entra-joined) device** for Microsoft 365, **require MFA**, **block legacy authentication**, and (for mobile) **require app protection** — all piloted in Report-only mode before enforcement, with break-glass accounts excluded.

The standard deployment model

Assess → Configure → Pilot → Validate → Expand.

Outcome

Zero Trust access enforcement that only admits healthy, authenticated devices and users — deployed safely, without locking anyone out.

1. Solution Overview — Conditional Access Integration

What Conditional Access is

Conditional Access (CA) is Microsoft Entra's policy engine that evaluates each sign-in and decides whether to **allow**, **challenge** (e.g. require MFA), or **block** it. Policies combine **signals** (user/group, target app, device state, location, client app, sign-in risk) with **access controls** (require MFA, require compliant

device, require Entra-joined device, require app protection policy, or block).

How it integrates with Intune

Intune marks devices **Compliant** (via device compliance policies) and applies **app protection policies** to mobile apps. Conditional Access consumes those signals: "**require a compliant device**" means only devices Intune has verified as healthy can reach Microsoft 365 — and "**require app protection policy**" means only managed mobile apps can access corporate data.

How it works

1. A user attempts to sign in to a protected app (e.g. Outlook).
2. Entra evaluates the applicable CA policies against the sign-in's signals.
3. If controls are satisfied (compliant device, MFA passed), access is granted; otherwise the user is challenged or blocked.
4. **Report-only** mode evaluates policies and logs what *would* happen without enforcing — the safe way to pilot.

The baseline policies this delivers

- **Require compliant or Entra-joined device** for Microsoft 365.
- **Require MFA** for all users.
- **Block legacy authentication** (the channel that bypasses MFA).
- **Require app protection policy** for mobile email/apps.
- **Break-glass exclusions** so emergency-access accounts are never locked out.

Benefits

- **Enforcement** — compliance and MFA become hard requirements, not suggestions.
- **Phishing/legacy resistance** — block legacy auth and require MFA.
- **Safe rollout** — Report-only mode and break-glass accounts prevent lockouts.
- **Zero Trust** — access decisions based on real-time device and identity signals.

Limitations (be honest with buyers)

- Conditional Access requires **Microsoft Entra ID P1**.
- "Require compliant device" depends on the **Device Compliance** solution being in place.
- Misconfigured policies *can* lock users (or admins) out — hence break-glass accounts and Report-only piloting.
- Some controls (app protection) require the managed mobile apps and enrollment posture.

The standard deployment model

> ***Create break-glass accounts → build policies in Report-only → review impact → enable → expand.*** *repeatable pattern:* ***Assess → Configure → Pilot → Validate → Expand.***

2. Requirements Guide — Conditional Access Integration

Verify **every** requirement before configuring. Each entry: **Description · Why · Verify · Implement-if-missing.** Category-wide prerequisites are in `global-requirements-intune-solutions.md`.

2.1 Licensing

Description: Microsoft Entra ID P1 (or P2 for risk-based policies). **Why:** Conditional Access is a P1 feature; risk-based conditions need P2. **Verify:** M365 admin center → **Billing** → **Licenses** (P1 via M365 E3/E5/Business Premium). **Implement:** Assign M365 E3/E5 or Business Premium to targeted users.

2.2 Break-glass (emergency access) accounts

Description: Two cloud-only Global Administrator accounts excluded from all CA policies. **Why:** They guarantee you can always sign in if a CA policy misfires — the #1 lockout safeguard. **Verify:** Entra → **Users** — two dedicated emergency accounts exist; run `CAI-BreakGlassCheck.ps1`. **Implement:** Create two cloud-only `.onmicrosoft.com` Global Admins with long random passwords stored offline; exclude them from every CA policy.

2.3 Device compliance (for "require compliant device")

Description: Intune device compliance policies in place. **Why:** The "require compliant device" control needs Intune to be marking devices Compliant. **Verify:** Intune → **Devices** → **Compliance**; devices report Compliant. **Implement:** Deploy the **Device Compliance Policies** solution first.

2.4 MFA registration

Description: Users are able to register MFA methods. **Why:** "Require MFA" challenges users; they must have a method registered. **Verify:** Entra → **Authentication methods**; run `CAI-Discovery.ps1` for the local user. **Implement:** Enable authentication methods and (optionally) the registration campaign or SSPR.

2.5 Sign-in logs / reporting

Description: Access to Entra sign-in logs. **Why:** Report-only impact is reviewed in the sign-in logs before enforcing. **Verify:** Entra → **Monitoring** → **Sign-in logs** loads. **Implement:** Ensure the operator has Reports Reader / Security Reader.

2.6 App protection (optional — mobile)

Description: Intune app protection policies for mobile apps. **Why:** The "require app protection policy" control needs them defined. **Verify:** Intune → **Apps** → **App protection policies**. **Implement:** Create app protection policies for iOS/Android (optional companion work).

Checklist

- Entra ID P1 licensing assigned
- **Two break-glass admins created and excluded** — `CAI-BreakGlassCheck.ps1`
- Device compliance policies in place
- MFA methods enabled / users can register
- Sign-in log access for review
- Pilot Entra group created

Global Requirements — Intune Solutions (category-level)

These requirements apply to **every** solution in the **Intune Solutions** category. They are published once at the category level so individual listings stay small. Confirm these before deploying *any* Intune solution. Each entry: **Description · Why · Verify · Implement-if-missing.**

G1. Microsoft Entra ID tenant

Description: An active Microsoft Entra ID (Azure AD) tenant. **Why:** It is the identity backbone for every cloud and Intune-managed resource. **Verify:** Sign in to `https://entra.microsoft.com` as an administrator. **Implement:** Create a Microsoft 365/Azure subscription, which provisions a tenant.

G2. Microsoft Intune subscription

Description: An Intune-bearing license plan. **Why:** Intune delivers all configuration, compliance, and security policies. **Verify:** `https://intune.microsoft.com` loads; **Tenant administration** → **Tenant status** shows active. **Implement:** Assign Microsoft 365 E3/E5, Business Premium, or standalone Intune Plan 1.

G3. Administrative roles

Description: Appropriate Entra/Intune admin roles for the operator. **Why:** Creating policies and enrollment settings requires elevated rights. **Verify:** Entra → **Roles and administrators**; confirm **Intune Administrator**. **Implement:** Assign **Intune Administrator**; reserve Global Administrator for tenant-wide changes.

G4. Device enrollment in Intune

Description: Targeted Windows devices are enrolled and manageable. **Why:** Policies only reach enrolled devices. **Verify:** Intune → **Devices** → **All devices**; the device appears and is **compliant**. **Implement:** Enable **Automatic Enrollment** and join devices to Entra ID.

G5. Pilot security group

Description: A Microsoft Entra security group for pilot/targeted users or devices. **Why:** Every solution is piloted before fleet rollout. **Verify:** Entra → **Groups**; the pilot group exists. **Implement:** Entra → **Groups** → **New group** → **Security** → add members.

G6. Supported Windows baseline

Description: Windows 10 21H2+ or Windows 11 with current updates. **Why:** Modern Intune features assume a patched, supported OS. **Verify:** `winver`, or Intune → device → **Hardware**. **Implement:** Deploy **Update rings** or upgrade to Windows 11.

G7. Network access to Microsoft cloud endpoints

Description: Outbound HTTPS to Microsoft 365, Entra, and Intune service URLs. **Why:** Enrollment, policy sync, and authentication depend on these endpoints. **Verify:** From a device, reach `https://login.microsoftonline.com` and `https://enrollment.manage.microsoft.com`. **Implement:** Allow Microsoft's published Office 365/Intune URL & IP ranges through proxy/firewall.

The standard deployment model (all solutions): Assess → Configure → Pilot → Validate → Expand.

3. Implementation Guide — Conditional Access Integration

Follow these steps **in order**. Conditional Access is in **Microsoft Entra** (`https://entra.microsoft.com` → **Protection** → **Conditional Access**). **Always start in Report-only.**

> ***Model:*** *Break-glass first → policies in Report-only → review → enable → expand.*

Step 1 — Create and exclude break-glass accounts

1. Entra → **Users** → **New user** → create two cloud-only `breakglass1@…onmicrosoft.com` and `breakglass2@…onmicrosoft.com` Global Administrators with long random passwords (store offline).
2. You will **exclude these two accounts from every CA policy** in the steps below.

Step 2 — Create the pilot group

1. Entra → **Groups** → **New group** → Security → `CA-Pilot` → add pilot users. **Create**.

Step 3 — Policy: require compliant device for Microsoft 365

1. **Protection** → **Conditional Access** → **Create new policy.**
2. **Name:** `CA - Require Compliant Device - M365`.
3. **Users:** Include `CA-Pilot`; **Exclude** the two break-glass accounts.
4. **Target resources:** Cloud apps → **Office 365**.

5. **Grant:** Grant access → **Require device to be marked as compliant** (and/or **Require Microsoft Entra hybrid joined device**); **Require all the selected controls**.

6. **Enable policy: Report-only**. Click **Create**.

Step 4 — Policy: require MFA

1. **Create new policy** → **Name:** `CA - Require MFA - All Apps`.

2. **Users:** Include `CA-Pilot`; **Exclude** break-glass accounts.

3. **Target resources:** All cloud apps.

4. **Grant: Require multifactor authentication**.

5. **Enable policy: Report-only**. **Create**.

Step 5 — Policy: block legacy authentication

1. **Create new policy** → **Name:** `CA - Block Legacy Auth`.

2. **Users:** Include `CA-Pilot`; **Exclude** break-glass accounts.

3. **Target resources:** All cloud apps. **Conditions** → **Client apps** → select **Exchange ActiveSync clients** and **Other clients**.

4. **Grant: Block access**.

5. **Enable policy: Report-only**. **Create**.

Step 6 — (Optional) require app protection for mobile

1. **Create new policy** → **Name:** `CA - Require App Protection - Mobile`.

2. **Users:** `CA-Pilot` (exclude break-glass). **Conditions** → **Device platforms** → iOS, Android.

3. **Grant: Require app protection policy** (and **Require approved client app**). **Report-only** → **Create**.

Step 7 — Review Report-only impact

1. Have pilot users sign in normally for a day or two.

2. Entra → **Monitoring** → **Sign-in logs** → open a sign-in → **Report-only** tab to see which policies *would* apply and the result.

3. Run `CAI-PolicyReport.ps1` and `CAI-SignInReport.ps1` to summarize impact. Resolve any unexpected blocks.

Step 8 — Enable enforcement

1. For each policy, edit **Enable policy** → **On**. Start with **Block Legacy Auth** and **Require MFA**, then **Require Compliant Device**.

2. Re-verify pilot users can work and break-glass accounts are excluded.

Step 9 — Expand to production

1. Change each policy's **Users** from `CA-Pilot` to **All users** (keeping break-glass exclusions).
2. Roll out in waves; monitor sign-in logs. Re-validate with `CAI-Validation.ps1`.

Order of operations

`Break-glass → Pilot group → Compliant-device (RO) → MFA (RO) → Block legacy (RO) → App protection (RO) → Review → Enable → Expand`

5. Testing Guide — Conditional Access Integration

Run during Report-only (Implementation Step 7) and again after enabling enforcement.

Test 1 — Policies evaluate in Report-only

Do: Entra → **Monitoring** → **Sign-in logs** → open a pilot sign-in → **Report-only** tab. **Expected:** Each policy shows a result (Would apply / satisfied / would block) — proving evaluation works.

Test 2 — Compliant device satisfies the control

Do: From a compliant pilot device, sign in to `https://outlook.office.com`. Run `CAI-Validation.ps1`. **Expected:** Access granted; the "require compliant device" policy is satisfied.

Test 3 — Noncompliant device is challenged/blocked

Do: From a noncompliant or unmanaged device, attempt the same sign-in (Report-only shows the would-be result). **Expected:** Report-only logs "would block"; after enforcement, access is blocked.

Test 4 — MFA is required

Do: Sign in as a pilot user from a new session. **Expected:** An MFA challenge appears (after enforcement) / Report-only logs "would require MFA".

Test 5 — Legacy authentication is blocked

Do: Attempt a legacy-auth connection (e.g. older IMAP client) as a pilot user. **Expected:** Report-only logs "would block"; after enforcement, the legacy client is denied.

Test 6 — Break-glass accounts are exempt

Do: Run `CAI-BreakGlassCheck.ps1`; sign in with a break-glass account. **Expected:** Break-glass accounts are excluded from all policies and sign in without CA controls.

Test 7 — Policy + sign-in report

Do: Run `CAI-PolicyReport.ps1` and `CAI-SignInReport.ps1`. **Expected:** A clear list of policies/states and a summary of CA results across recent sign-ins.

Pass/fail gate

Gate	Requirement
Pilot pass (Report-only)	Tests 1–6 behave as expected with **no unexpected** would-blocks for legitimate users
Enforcement pass	After enabling, pilot users work normally; break-glass verified; no lockouts

6. Troubleshooting Guide — Conditional Access Integration

Use `CAI-Validation.ps1` / `CAI-SignInReport.ps1` to see why a sign-in was controlled, and `CAI-BreakGlassCheck.ps1` before any enforcement change.

A. Lockout risk / admins blocked

Symptom	Cause	Remediation
Admin locked out by a policy	Break-glass not excluded, or policy too broad	Sign in with a **break-glass** account; fix the exclusion; never enforce without break-glass verified
Everyone blocked after enabling	Policy enabled with a misconfigured grant	Use break-glass to set the policy back to **Report-only**; review the grant controls

B. "Require compliant device" blocks legitimate users

Symptom	Cause	Remediation
Compliant-looking device blocked	Device not actually Compliant in Intune, or not Entra-registered	Run `CAI-Validation.ps1`; confirm Intune shows **Compliant**; fix compliance (DCP solution)
Browser blocked but app works	Device registration/SSO state differs per client	Ensure the device is Entra-registered/joined and the browser uses the work account

C. MFA issues

- Users repeatedly prompted → expected on new sessions; confirm "remember MFA"/sign-in frequency settings.
- Users with no method → enable authentication methods and have them register before enforcing.

D. Legacy auth still working

- Block-legacy policy in Report-only → it only logs. Move to **On** to enforce.
- Some "modern-looking" clients still use legacy endpoints → confirm via sign-in logs **Client app** column.

E. Report-only shows nothing

- No sign-ins yet for the targeted users/apps → have pilot users actually sign in; wait for logs to populate (a few minutes).

F. Diagnostics quick reference

```
dsregcmd /status        # device join/registration state (affects compliant-device control)
```

Entra → **Sign-in logs** → **(a sign-in)** → **Conditional Access / Report-only** tabs show exactly which policies applied and why. The **What If** tool (Conditional Access → What If) simulates a sign-in.

G. When to escalate

If a sign-in is blocked with all controls apparently satisfied (device Compliant, MFA passed) and the Report-only/CA tabs are inconsistent, capture the sign-in correlation ID and open a Microsoft support case.

7. Best Practices — Conditional Access Integration

- **Create break-glass accounts first.** Two cloud-only Global Admins, excluded from every policy, passwords stored offline. This is the single most important safeguard against lockout.
- **Always pilot in Report-only.** Never enable a new policy straight to On — review the would-be impact in sign-in logs first.
- **Exclude break-glass from every policy.** Make it a checklist item on each policy you create.
- **Block legacy authentication.** It's the channel attackers use to bypass MFA — high value, low risk to enable.
- **Layer the controls.** Require MFA + require compliant device + block legacy auth together form a strong baseline.
- **Use the What If tool.** Simulate a user/app/condition before enabling to predict the outcome.
- **Roll out by waves.** Move from pilot group to all users gradually, watching sign-in logs.
- **Name policies clearly.** `CA - <control> - <scope>` so intent is obvious and audits are easy.
- **Monitor continuously.** Review sign-in logs and `CAI-SignInReport.ps1` for unexpected blocks after each change.
- **Pair with compliance.** "Require compliant device" is only as strong as your Device Compliance policy — deploy that solution alongside.

8. Deliverables Summary — Conditional Access Integration

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	What Conditional Access is, how it integrates with Intune, benefits, limitations
2. Requirements Guide	Every prerequisite with Verify / Implement-if-missing
Global Requirements (Intune Solutions)	Category-wide prerequisites, published once
3. Implementation Guide	Exact, ordered steps — Report-only first, break-glass safety
5. Testing Guide	Validation procedures with pass/fail gates
6. Troubleshooting Guide	Field issues and lockout recovery
7. Best Practices	Field-proven recommendations

PowerShell scripts

Script	What it does
`CAI-Discovery.ps1`	Local device/identity state: Entra join, registration, MFA-relevant facts
`CAI-Validation.ps1`	Confirms the device satisfies "require compliant/joined device" controls
`CAI-PolicyReport.ps1`	Lists Conditional Access policies and their state (Microsoft Graph)
`CAI-SignInReport.ps1`	Summarizes recent sign-ins and their CA results (Microsoft Graph)
`CAI-BreakGlassCheck.ps1`	Verifies break-glass accounts are excluded from all CA policies

What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process following the repeatable model: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

Zero Trust access enforcement — only healthy, authenticated devices and users get in — deployed safely with break-glass accounts and Report-only piloting.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

CAI-BreakGlassCheck.ps1

Purpose: For the break-glass accounts you specify, confirms each Conditional Access policy excludes them (directly or via an excluded group). The single most important pre-enforcement check. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\CAI-BreakGlassCheck.ps1 -Upns breakglass1@contoso.onmicrosoft.com,breakglass2@contoso.onmicrosoft.com Requires Microsoft Graph SDK and Policy.Read.All + User.Read.All + GroupMember.Read.All. EXPECTED RESULTS Per policy: EXCLUDED (good) or NOT EXCLUDED (fix before enforcing). .NOTES White-label — Intune Solutions. Read-only against Graph. #> [CmdletBinding()] param([Parameter(Mandatory)][string[]]$Upns) if (-not (Get-Module -ListAvailable -Name Microsoft.Graph.Authentication)) { Install-Module Microsoft.Graph -Scope CurrentUser -Force -AllowClobber } Import-Module Microsoft.Graph.Authentication -ErrorAction Stop Connect-MgGraph -Scopes 'Policy.Read.All','User.Read.All','GroupMember.Read.All' -NoWelcome

How to run: powershell -ExecutionPolicy Bypass -File .\CAI-BreakGlassCheck.ps1 -Upns breakglass1@contoso.onmicrosoft.com,breakglass2@contoso.onmicrosoft.com Requires Microsoft Graph SDK and Policy.Read.All + User.Read.All + GroupMember.Read.All. EXPECTED RESULTS Per policy: EXCLUDED (good) or NOT EXCLUDED (fix before enforcing). .NOTES White-label — Intune Solutions. Read-only against Graph. #> [CmdletBinding()] param([Parameter(Mandatory)][string[]]$Upns) if (-not (Get-Module -ListAvailable -Name Microsoft.Graph.Authentication)) { Install-Module Microsoft.Graph -Scope CurrentUser -Force -AllowClobber } Import-Module Microsoft.Graph.Authentication -ErrorAction Stop Connect-MgGraph -Scopes 'Policy.Read.All','User.Read.All','GroupMember.Read.All' -NoWelcome

CAI-Discovery.ps1

CAI-PolicyReport.ps1

CAI-SignInReport.ps1

CAI-Validation.ps1